Case G — Mr. Khan

28M · Software Engineer · 2-year progressive low back & buttock pain

Musculoskeletal Physiotherapy

■ SUBJECTIVE PROBLEM LIST

- Physical Impairments
- Reduced lumbar flexion (Schober's test <5cm)
- Restricted chest expansion (2cm — severely reduced)
- Inflammatory pain at rest and at night
- Fatigue (common in active inflammatory disease)
- Difficulty sustaining static postures at desk (prolonged sitting as software engineer)
- Reduced walking tolerance during flares
- Difficulty with bending/dressing secondary to spinal stiffness
- Sedentary software engineering role with prolonged static postures may perpetuate stiffness and deconditioning.
- 2-year history may have caused anxiety, fear-avoidance, or loss of confidence in movement — worth screening with a validated tool (e.g. STarT Back).

■■ RED FLAGS

- ■ Saddle Anaesthesia / Bowel-Bladder Change
- Significance:
- Cauda equina syndrome — surgical emergency
- Immediate emergency referral to A&E; cease assessment
- ■ Unexplained Weight Loss / Fever / Night Sweats
- Potential malignancy or systemic infection (e.g. spinal TB)
- Urgent same-day GP/oncology referral; imaging required
- ■ History of Malignancy

■ PATHOPHYSIOLOGY (KEY POINTS)

- Inflammatory cytokines, particularly TNF-α and IL-17, promote synovitis, bone marrow oedema, and ultimately osteoproliferation: the pathological formation of new bone (syndesmophytes) at the vertebral endplates.
- The SIJ is typically affected first bilaterally, distinguishing it from mechanical back pain, and costovertebral involvement restricts chest expansion below the normal threshold of ≥5cm.
- Inflammatory nocturnal pain (4am waking): Pro-inflammatory cytokines peak in the early hours of the morning, correlating with symptom worsening at rest when the anti-inflammatory effect of cortisol is lowest.
- Morning stiffness >1 hour, improved by movement: Inflammatory exudate within SIJ/spinal joints accumulates during rest; movement promotes synovial fluid circulation and reduces joint stiffness.
- Bilateral buttock pain: Bilateral SIJ enthesitis and synovitis refers pain to the gluteal region — often described as alternating sides in early disease.
- Reduced lumbar flexion (Schober's test): Spinal and paraspinal enthesitis with early syndesmophyte formation restricts segmental mobility.

■ DIFFERENTIAL DIAGNOSES

1. No history of uveitis or systemic features
2. Another seronegative spondyloarthropathy
3. Classic anterior uveitis is more typical of AS than PsA

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

■ Key Observation Findings (State These Before Tests)

Posture: Observe for flattening of lumbar lordosis, increased thoracic kyphosis, and forward head posture — early signs of spinal ankylosis in AS Gait: Rigid, stiff lumbar gait with reduced spinal rotation — distinguishes from antalgic gait of discogenic pain Chest expansion:

■ TREATMENT PLAN

- ■ CONTRAINDICATION — HVLA Spinal Manipulation

■ Goal:

- Hydrotherapy / Pool-Based Exercise:
- Warm water reduces muscle guarding and pain, enabling active movement in an anti-gravity environment — ideal for active inflammatory disease.
- — I choose this because the warmth and buoyancy allow exercise that would be painful or impossible on land during flares.
- Breathing Exercises:
- Diaphragmatic and thoracic expansion exercises to maintain chest wall mobility given 2cm chest expansion.
- — I prioritise this because costovertebral ankylosis is progressive and irreversible; early intervention preserves respiratory function.
- Gentle Sustained Stretching (not ballistic):
- Hip flexor and thoracic rotation stretches held 30–60 seconds.
- — Sustained low-load stretching is preferred over aggressive mobilisation given tissue inflammation; it gently lengthens without provocative loading.
- Exercise 1: Thoracic Extension Over Foam Roller